

Predicting K-POP Song Popularity:

A comparative study of regression, classification, company-level analysis

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ABSTRACT

This paper analyzes the factors driving the commercial success of K-Pop songs using a combination of audio features, lyrics, and agency data. We implemented both regression and classification pipelines to predict chart rankings and hit status (Top 35). Our experiments show that intrinsic audio features are insufficient for predicting exact rankings, as evidenced by near-zero R^2 scores across all regression models. In contrast, predicting whether a song becomes a "hit" yielded better results, with Logistic Regression achieving 62% accuracy. A key finding is the dominance of the "Company" feature, indicating that a song's success in the K-Pop market is heavily dependent on influence of the management agency rather than musical attributes alone.

Keywords K-Pop, Music Popularity Prediction, Machine Learning, Spotify Audio Features, Hit Song Science, Entertainment Agency Analysis, Data Science

1. INTRODUCTION

K-Pop has emerged as a major force in the global music industry in recent years. Our study is driven by the desire to empirically investigate the factors that determine a K-Pop song's success, particularly in light of the widely discussed notion of K-Pop being produced on a "factory assembly line." Specifically, we aim to determine if a formula for guaranteed success can be quantified. To address this, we leverage various song-idol features to predict whether a song is popular.

2. METHOD

Our methodology involved constructing a predictive model using multi-dimensional feature sets to investigate the mechanisms of K-Pop song success. We utilized various inputs, including objective Audio Features sourced from Spotify (such as Danceability, Energy, and Tempo), Lyric Data (the textual content of the songs), and Idol and Agency Features (information related to the artists' management companies). With these features, we pursued two primary predictive objectives: first, to perform quantitative ranking prediction by forecasting the song's specific chart rank, and second, to execute popularity classification by predicting whether the song would be considered a hit (i.e., entering the Top 35 rankings).

3. EXPERIMENTS

3.1 Dataset / Environment

The datasets came from Kaggle and GitHub, covering audio features, lyrics, rank of songs, and group information such as gender and company. Due to the fragmented nature of the data sources, multiple datasets needed to be integrated. We achieved this by matching the artist name and song title against the Spotify API to retrieve the corresponding Track ID. This unique Track ID was then used as the primary key to merge the dataset of audio features with the dataset containing song lyrics and rank information. A critical step to maximize search accuracy was the normalization of artist names. We specifically converted Korean-script artist names into their

widely recognized English counterparts prior to using them in the Spotify search query.

3.2 Experimental Setup

The experimental pipeline was implemented using Python 3.10 with the Scikit-learn, XGBoost, and LightGBM libraries. For data preprocessing, we applied One-Hot Encoding to categorical features (e.g., Company, Gender) and normalized continuous audio features using StandardScaler. The dataset was partitioned into 80% training and 20% testing sets. Specifically for the classification task, stratified sampling was employed to ensure consistent class distribution across subsets.

To address the class imbalance caused by the scarcity of hit songs (Top 35), we applied SMOTE (Synthetic Minority Over-sampling Technique) to the training set. Finally, we evaluated the performance of multiple models, including Linear/Logistic Regression, Random Forest, XGBoost, and LightGBM, to compare their predictive capabilities on both ranking and classification tasks.

3.3 Evaluation Metrics

We evaluate the proposed method using standard regression and classification metrics. For regression tasks, Mean Squared Error (MSE) and the coefficient of determination (R^2) are adopted. For classification performance, accuracy and F1-score are reported.

3.4 Results

3.4.1 Performance of Ranking Prediction

The result clearly showed that exact ranking prediction is difficult, while hit classification is feasible. Performance of Ranking Prediction

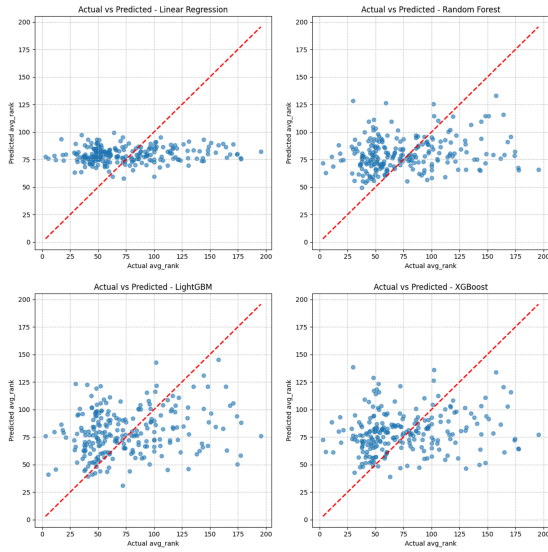
Table 1 summarizes the regression results. The models failed to model the variance in rankings effectively. Linear Regression yielded the only positive R^2 score (0.0058), while

tree-based models produced negative values. This lack of predictive power is visually confirmed in Figure 1, where the scatter plot shows a vertical distribution rather than the expected diagonal correlation.

Table 1: Performance of Regression Models

Model	MSE	MAE	R ²
Linear Regression	1471.9802	31.8449	0.0058
Random Forest	1479.6589	30.5514	0.0006
XGBoost	1599.0760	31.5336	-0.0801
LightBGM	1561.3749	31.1130	-0.0546

Figure 1: Actual vs. Predicted Rankings (Linear Regression, Random Forest, XGBoost, and LightGBM respectively)



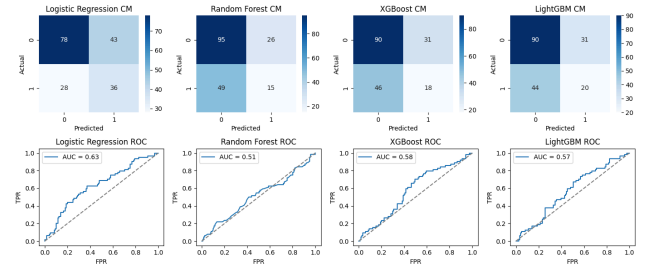
3.4.2 Performance of Hit Classification

In contrast, the binary classification task (Top 35 prediction) yielded more promising results. As shown in Table 2, Logistic Regression achieved the highest accuracy of 62%. Figure 2 presents the Confusion Matrices and ROC Curves, confirming that the models perform statistically better than random guessing (AUC > 0.5).

Table 2: Performance of Classification Models (Target: Top 35)

Model	Accuracy	Precision	Recall	F1-score
Logic Regression	0.62	0.64	0.62	0.62
Random Forest	0.59	0.56	0.59	0.57
XGBoost	0.58	0.56	0.58	0.57
LightGBM	0.59	0.57	0.59	0.58

Figure 2: Classification Performance.(upper) Confusion Matrices; (lower) ROC Curves showing the trade-off between sensitivity and specificity.



To gain insight into the factors driving the hit prediction, we analyzed the top 10 feature importances for all four classification models.

LightGBM, Random Forest favor continuous, granular audio features (e.g., energy, duration_ms). In contrast, Logistic Regression and XGBoost models assign higher importance to discrete categorical features (e.g., company and gender), indicating the strong influence of the music industry landscape on commercial outcomes.

Figure 3(a): Top 10 important features of Classifier using Logistic Regression

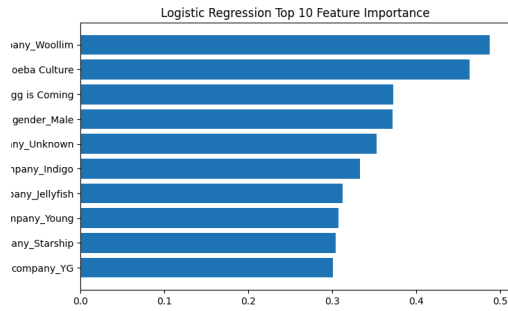


Figure 3(b): Top 10 important features of Classifier using XGBoost

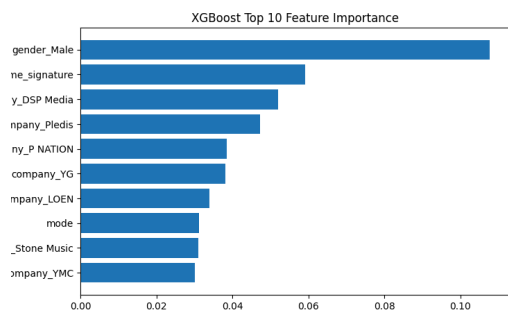


Figure 3(c): Top 10 important features of Classifier using Random Forest

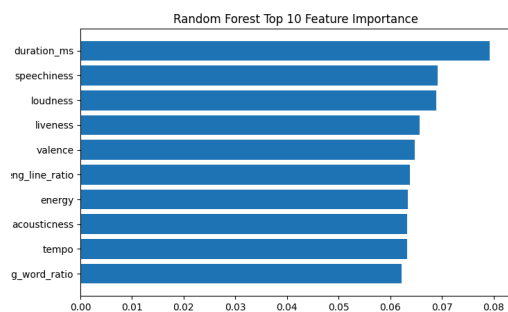
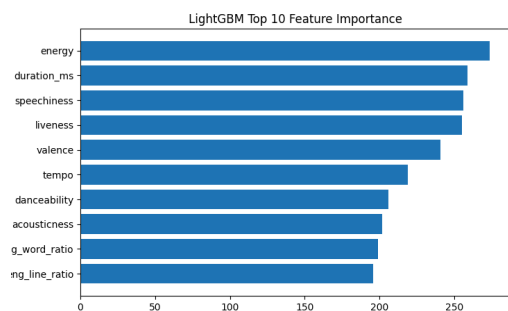


Figure 3(d): Top 10 important features of Classifier using LightGBM

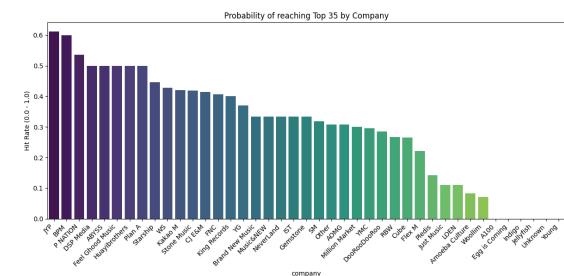


3.4.2 Company-Level Analysis

Building on the findings from the feature importance analysis, where company identity emerged as a significant predictor for Logistic Regression and XGBoost models, we conducted a detailed examination of the "hit rate" (probability of reaching the Top 35) across different agencies.

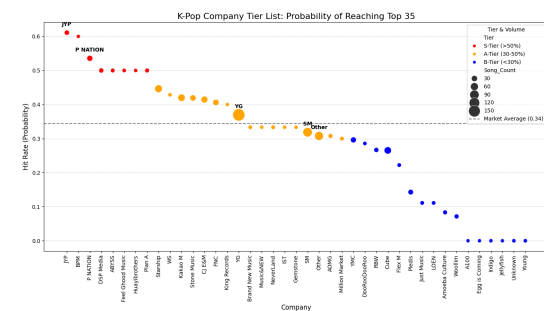
Figure 4(a) illustrates a significant disparity in market performance among agencies. Notably, JYP Entertainment and P NATION lead the industry with hit rates exceeding 60% and 50%, respectively. However, raw probability alone can be misleading, as smaller agencies with few releases may show artificially high success rates due to small sample sizes.

Figure 4(a): probability of reaching the Top 35 among different agencies



To address this variance in release frequency through a volume-weighted tier analysis, Figure 4(b) categorizes agencies into tiers based on hit probability while simultaneously visualizing their output volume (represented by bubble size).

Figure 4(b): Volume-Weighted Agency Performance Tiers



4. DISCUSSION/ANALYSIS

4.1 Limitations of Regression Models

The experimental results for the regression task were suboptimal. Linear Regression yielded an R^2 score of only 0.0058, while XGBoost and LightGBM produced negative R^2 scores. This suggests that Spotify audio features (e.g., loudness, tempo) alone are insufficient to predict the exact numerical ranking of a song. The variance in rankings is likely driven by complex external factors—such as fandom dynamics, marketing timing, and viral trends—that cannot be captured linearly by audio data.

4.2 Efficacy of Classification

When the problem was simplified to a binary classification task (Top 35 vs. Others), the model performance improved. Logistic Regression achieved an accuracy of 62%, which is statistically better than random guessing. This indicates that while exact ranking is unpredictable, there are distinguishable patterns that separate "hit songs" from the rest.

4.3 Importance of the "Company" Feature

A key finding from our Feature Importance Analysis (visualized in [company_feature/prob_of_hit_by_company.py](#)) is the dominance of the "Company" variable. In several models, the agency associated with the artist was a stronger predictor than the audio features. This validates the hypothesis that in the K-Pop industry, the resources and brand equity of the agency are major determinants of a song's commercial success.

5. CONCLUSION

Our comparative study concludes that intrinsic audio features are insufficient for predicting K-Pop success. The structural limitations of the regression models and the moderate success of the classification models highlight that the K-Pop market is driven heavily by external factors. The most significant predictor identified was the management agency, with specific companies showing distinct "hit rates." Future work should

focus on integrating data related to marketing efforts, such as Music Video promotion quality and social media fan engagement, to build a more accurate predictive model.

6. REFERENCES

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